6-2.

一维, $D(p)dp = \frac{L}{h}dp$.

又 $p = \sqrt{2\varepsilon m}$, $dp = \sqrt{2m}\cdot\frac{1}{2}\frac{1}{\sqrt{\varepsilon}}d\varepsilon = \sqrt{\frac{m}{2\varepsilon}}d\varepsilon$

$D(\varepsilon)d\varepsilon = \frac{L}{h}\sqrt{\frac{m}{2\varepsilon}}d\varepsilon$.

6-3. $D(p)dp = \frac{1}{h^2}dp_x dp_y dq_x dq_y = \frac{L^2}{h^2}dp_x dp_y$

$= \frac{L^2}{h^2}p\,dp\,d\theta$. $p = \sqrt{2\varepsilon m}$ $dp = \sqrt{\frac{m}{2\varepsilon}}d\varepsilon$

$d\theta = 2\pi$

$= \frac{2\pi L^2}{h^2}\cdot\sqrt{2\varepsilon m}\cdot\sqrt{\frac{m}{2\varepsilon}}d\varepsilon$

$= \frac{2\pi m L^2}{h^2}d\varepsilon$

6-4. $p = \frac{\varepsilon}{c}$. $dp = \frac{1}{c}d\varepsilon$.

$\frac{1}{h^3}\cdot V\cdot dp_x dp_y dp_z = \frac{V}{h^3}p^2\sin\theta\,dp\,d\theta\,d\varphi = \frac{4\pi V}{h^3}p^2dp$

(θ: $0 \to \pi$; φ: $0 \to 2\pi$; ~~$-\cos\theta$~~)

$= \frac{4\pi V}{h^3}\frac{\varepsilon^2}{c^3}\cdot\frac{1}{c}d\varepsilon = \frac{4\pi V\varepsilon^2 d\varepsilon}{h^3c^3}$

总结：

ε.

$$\frac{1}{h^i}\, dp_1\, dp_2 \cdots dp_i\, dq_1\, dq_2 \cdots dq_i = \ ?$$

线度 $\begin{cases} \text{体积 } V \\ \text{长度 } L \\ \text{面积 } S. \end{cases}$